Yi Cao

AI4Science | Computer-Aided Material Design | Machine Learning | LLMs | Explainable AI | Scientific Computing

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Professional Summary

I am a PhD researcher building trustworthy, physics-grounded AI for materials discovery. I develop frameworks that bridge large scale molecular dynamics with first-principles simulations by machine-learned force fields, spanning explainable AI (AAAI 2026 XAI4Science), generalization benchmarking (non-equilibrium probes revealing specialist vs. generalist trade-offs, NeurIPS 2025 AI4Mat spotlight), and causal reasoning (KDD 2026). Recognized with Best Poster awards across multiple venues; invited as session chair at PHM 2025. Strong foundation in both AI/ML methodology and computational materials science, with demonstrated experience translating research into industry applications.

Education

PhD **Johns Hopkins University**, Chemical and Biomolecular Engineering (ChemBE) — Aug. 2023 to May 2028 (Expected)

- GPA: 4.0/4.0
- **Advisor:** Paulette Clancy
- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Modern Data Analysis, Databases, Algorithms, Computing for Applied Mathematics, Statistical Mechanics

MS **Johns Hopkins University**, Computer Science — Aug. 2026 to May 2028 (Expected)

- **Focus:** Machine learning, AI4Science, Agentic AI

BS **Fudan University**, Pharmaceutical Sciences — Sept. 2019 to July 2023

- GPA: 3.7/4.0 (Rank: 2/112)
- **Awards:** "Graduate Star" Nomination Award (2023), Excellent graduates of Shanghai Colleges and Universities (2023), 1st Class Scholarship (2021), Shanghai Scholarship (2020)

University of California, Berkeley, Visiting Scholar — Aug. 2021 to Dec. 2021

Research Experiences

Johns Hopkins University, Ph.D. Research under Advisor Paulette Clancy — MD, USA, Nov. 2023 to present

- Developing AI frameworks that integrate molecular simulation (DFT, MD) with machine learning and LLMs for 2D materials discovery and design
- Created **dual explainability** framework for machine-learned force fields (MLFFs), bridging model reasoning with human understanding (*XAI4Science-AAAI-2026*).
- Developed a **causal-aware KG–LLM** integration framework that mitigates “contextual tunneling” and improves scientific reasoning (*ICLR 2026 submission*).
- Benchmarked specialist vs. generalist **MLFFs** using migration pathways as diagnostic probes, revealing task-specific fine-tuning trade-offs (*AI4Mat-NeurIPS-2025, spotlight talk*).
- **Skill set**: Machine Learning (MLFFs, LLMs, causal reasoning, transfer learning), Molecular Simulation (DFT, MD), Scientific Computing

Institute of Science and Technology for Brain-inspired Intelligence, Fudan University, Undergraduate Research — Shanghai, China, Feb. 2022 to Jun. 2023, 16 months

- Utilized **data mining and signal processing** on large-scale brain recordings to uncover disease-related patterns, improving diagnostic interpretation.

- **Skill set**: Data mining, Brain-signal processing, Neuropharmacology

School of Engineering, Westlake University, Summer Research

Hangzhou, China
July 2022 to Aug. 2022
2 months

- Applied **ML-driven polymer screening** to guide nanoparticle design for brain-targeted drug delivery, linking material optimization with experimental validation.
- **Skill set**: Machine Learning for materials design, Nanoparticle characterization

Technical Skills

Machine Learning & AI: Deep Learning (PyTorch, TensorFlow), Large Language Models (fine-tuning, prompt engineering), Causal Inference, Graph Neural Networks, Transfer Learning, Explainable AI
Scientific Computing: Molecular Dynamics (LAMMPS, GROMACS), Density Functional Theory (Quantum ESPRESSO), High-Performance Computing (MPI, CUDA)
Programming & Tools: Python, MATLAB, R, Git, Docker, Linux/Unix
Data Science: Large-scale data processing, Statistical analysis, Knowledge graphs, Database management (SQL), Distributed computing
Specialized: Materials informatics, Scientific AI

Publications

1. **RepliCan: Evaluating LLM Agents on Scientific Reproducibility in Computational Materials Science**
   Ziyang Huang, ***Yi Cao***, Ali K. Shargh, Jing Luo, Ruidong Mei, Mohd Zaki, Zhan Liu, William Jurayj, Somdatta Goswami, Michael Shields, Jaafar El-Awady, Paulette Clancy, William Gantt Walden, Nicholas Andrews, Benjamin Van Durme, Daniel Khashabi
   Accepted to COLM Conference (2026)

2. **Atomic Switch Control via Two-Mode Intercalation for Tunable 2D Materials**
   Yi Cao, Victor Wu, and Paulette Clancy*
   npj 2D Materials and Applications, under review (2025)

3. **What is Your Force Field *Really* Learning? Gaining Scientific Intuition with A Dual-Level Explainability Framework**
   Yi Cao, Peter Mastracco, Jieneng Chen, Alan Yuille, Paulette Clancy*
   AAAI Conference Workshop (XAI4Science), Peer-reviewed Workshop Paper (2026)
   **Dual-level explainability framework bridging model reasoning with human understanding in scientific AI*

4. **ARIA: A Causal-Aware Framework for Rescuing LLM Reasoning in Trustworthy Materials Discovery**
   Yi Cao, Liaoyaqi Wang, Jieneng Chen, Benjamin Van Durme, Alan Yuille, Paulette Clancy*
   Submitted to KDD Conference, AI4Sciences Track (2026)

5. **Migration as a Probe: A Generalizable Benchmark Framework for Specialist vs. Generalist Machine-Learned Force Fields**
   Yi Cao and Paulette Clancy*
   NeurIPS Conference Workshop (AI4Mat), Peer-reviewed Workshop Paper (2025)
   **Comprehensive benchmarking framework for evaluating MLFF generalization in materials science*

6. **Low-energy pathways lead to self-healing defects in $CsPbBr_3$**
   Kumar Miskin, ***Yi Cao***, Madaline Marland, Farhan Shaikh, David T. Moore, John Marohn, Paulette Clancy*
   Phys. Chem. Chem. Phys., 27(29), 15446-15459 (2025)
   **Computational discovery of self-healing mechanisms with implications for rational material design*

7. **Brain-targeted nanoparticle drug delivery systems: research advances**
   Yi Cao and Chen Jiang*
   Basic and Clinical Medicine, 42(1), 2–14 (2022)

Conference Presentations

2026 MRS Spring Meeting & Exhibit (Honolulu, HI, May 2026) - ***Oral Presentation***
Migration as a Probe—A Generalizable Benchmark Framework for Specialist vs. Generalist Machine-Learned Force Fields in Doped Materials

2026 MRS Spring Meeting & Exhibit (Honolulu, HI, Apr 2026) - ***Oral Presentation***
What is Your Force Field Really Learning? Gaining Scientific Intuition and Guiding Feature Engineering with A Dual-Level Explainability Framework

2026 MRS Spring Meeting & Exhibit (Honolulu, HI, Apr 2026) - ***Oral Presentation***
Overcoming Contextual Tunneling in AI-Driven Materials Discovery: A Causal Framework for Robust Knowledge Integration
Women in Data Science and AI Symposium (Baltimore, MD, Apr 2026) - ***Best Poster Award Winner***

Women of Whiting Annual Women in STEM Symposium (Baltimore, MD, Mar 2026) - ***Top 1 Poster Presentation Award***

AAAI XAI4Science Workshop (Singapore Expo, Jan 2026) - ***Spotlight Talk*** (Top-tier recognition)

What is Your Force Field Really Learning? Gaining Scientific Intuition with A Dual-Level Explainability Framework

NeurIPS AI4Mat Workshop (San Diego, CA, Dec 2025) - ***Spotlight Talk & Travel Grant Awardee*** (Top-tier recognition)
Migration as a Probe: A Generalizable Benchmark Framework for Specialist vs. Generalist Machine-Learned Force Fields

Prognostics and Health Management (PHM) Society Annual Conference (Bellevue, WA, Oct 2025) - ***Invited Talk & Session Chair***
Doctoral Symposium - Selected as 1 of 10 PhD students globally; showcased AI applications in predictive maintenance and materials health; Chaired "Advanced AI/ML for PHM" and "Digital Twins and Aviation" sessions

AIChE Annual Meeting (Boston, MA, Nov 2025)
Poster: Dissecting Cr-Doped Interfaces: How Diffusion Dynamics Reshape the Van Der Waals Gap and Shift Sb_2Te_3 Thermoelectric Performance

Women in AI 2025 (Baltimore, MD, Apr 2025) - ***Poster Presentation Award Winner***
Exploring Low-Energy Pathways for Self-Healing Defects in CsPbBr3—A Computational Study

Materials Research Society (MRS) Fall Meeting (Boston, MA, Dec 2024)
Poster: Exploring Low-Energy Pathways for Self-Healing Defects in $CsPbBr_3$

Additional Experiences And Awards

Qualcomm, Engineering Intern (GPU High-Level Modeling) — San Diego, CA, USA; May 2026 to Aug. 2026; Internship

- Developing an **agentic AI workflow** to automate complex debugging processes in GPU high-level modeling (HLM), integrating LLM capabilities with existing diagnostic tools to streamline operations.

LLM Hackathon for Applications in Materials Science & Chemistry — Global (JHU Site, MD, USA); Sept. 2025; Team Leader

- Led the **ARIA team** (Autonomous Reasoning Intelligence for Atomics), integrating LLMs with causal knowledge graphs for robust inverse materials design.
- Received a **2025 Visionary Award** (ranked 6th out of 120 teams).
- **Industry Recognition:** Project noticed by CuspAI, leading to ongoing collaboration discussions for commercial applications

Viva Biotech — Shanghai, China; Jun. 2024 to Jul. 2024; Internship

- Conducted Computer-Aided Drug Design (CADD) research using co-solvent MD simulations, optimizing drug discovery through protein-ligand interaction analysis.

Fudan iGEM team, iGEM Competition — Shanghai, China; Dec. 2022 to Nov. 2023; Part-time, Advisor

- Guided experimental design, scientific documentation.
- Our team won a **Gold Medal** and **Best Environmental Project** title.

Boehringer Ingelheim — Pudong, Shanghai, China; Dec. 2022 to Feb. 2023; Internship

- Led a team in developing a white paper on quality culture through research and interviews, resulting in improved company-wide quality guidelines.